

Ronak Dutta

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EDUCATION

University School of Automation and Robotics (GGSIPU), New Delhi

- Bachelor of Technology in Industrial Internet of Things (Expected September 2027) | GPA: 8.81/10
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database System, Computer Networks

Bal Mandir Sr. Sec. School, Delhi

- CBSE (Class XII) | Percentage: 92.6% (2022)

WORK EXPERIENCE

1M1B Green Internship – Smart Irrigation IoT System

July 2025 – Aug 2025

- Designed a prototype smart irrigation system using IoT devices (ESP32, soil moisture & temperature sensors) to automate water scheduling.
- Built a web-based dashboard for real-time monitoring and remote control of irrigation, enabling actionable water-use insights.
- Leveraged Tableau for sustainability dashboards, visualizing patterns and trends in environmental data.

PROJECTS

SkillSwap - Decentralized Skill Exchange (PostgreSQL, Express, React, Node.js, TailwindCSS)

[Live Demo](#)

- Architected a full-stack application enabling users to monetize or barter skills via a custom virtual currency system.
- Engineered a robust REST API with Node.js and PostgreSQL to handle complex transaction logic, including wallet deductions, skill ownership transfers, and atomic swap negotiations.
- Executed secure JWT authentication and a "Handshake Protocol" that allows users to initiate, accept, or reject barter requests with real-time status updates.

ATS Workplace - AI-Powered Recruitment Suite (React, Node.js, Python, FastAPI, PostgreSQL)

[Live Demo](#)

- Architected a microservice-based ATS featuring a React frontend, a Node.js API gateway, and a Python-based ML engine to automate high-volume recruitment workflows.
- Developed a high-performance NLP pipeline using FastAPI, SpaCy, and Hugging Face Transformers to perform real-time candidate analysis and semantic resume scoring against job descriptions
- Engineered an automated screening system that generates AI-driven candidate summaries and ranking metrics, significantly reducing manual review time for HR teams.

SafarSaathi - Cab Booking Application (PostgreSQL, Express, React, Node.js, Twilio, TailwindCSS)

[Live Demo](#)

- Implemented a multi-tier RBAC (Role-Based Access Control) system managing distinct authentication flows for Admins, Drivers, and Customers via JWT.
- Developed a comprehensive Admin Dashboard with data visualization for revenue tracking and fleet management system that allows real-time driver assignment, triggering automated WhatsApp confirmations via Twilio.
- Implemented a complete driver onboarding workflow - from application to approval - and integrated Razorpay for secure payment processing, streamlining the end-to-end booking lifecycle.

SKILLS & CERTIFICATIONS

TECHNICAL SKILLS

- **Programming Languages:** JavaScript (ES6+), TypeScript, Python, Java, C++, SQL
- **Front-End:** React.js, Next.js, Tailwind CSS, Redux
- **Back-End:** Node.js, Express.js, Axios
- **Databases:** MongoDB, PostgreSQL, MySQL, Firebase
- **Tools & Platforms:** Git, GitHub, Postman, Vercel

CERTIFICATIONS & TRAINING

- React – The Complete Guide (Hooks, React Router, Redux) – Jonas Schmedtmann (Udemy)
- The Complete Full-Stack Web Development Bootcamp – Dr. Angela Yu (Udemy)

LANGUAGES

- **English** – Fluent | **Hindi** – Native | **Bengali** – Native